

Sonam Okuda

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Education

University of Pennsylvania

M.S.E. in Mechanical Engineering and Applied Mechanics

May 2027

Philadelphia, PA

The Cooper Union for the Advancement of Science and Art

B.E. in Mechanical Engineering, Minor in Computer Science

May 2025

New York, NY

Full Tuition Scholarship (2024–2025); Innovator Tuition Merit (2021–2025)

Technical Skills

Programming & Software:	Python, C/C++, Linux/Ubuntu, ROS2, RViz, Foxglove, Git, Arduino, JS, HTML/CSS
Robotics & Controls:	Motion planning (A*, Dijkstra, RRT, PRM), forward/inverse kinematics, PID/LQR control, trajectory optimization
Sensing:	LiDAR, SLAM, particle filters, EKF/UKF, NeRF, GNNs, computer vision
Hardware:	Franka FR3 & Panda arms, CrazyFlie, Pixhawk, ESP32, Arduino Mega, sensor integration
Design & Fabrication:	Siemens NX, Onshape, Solidworks, 3D printing, laser cutting, soldering, CNC routing, prototyping and testing

Experience

GRASP Lab: Autonomous Manipulation @ Penn

Mar 2026 – Present

Student Researcher, advised by Dr. Rachel Holladay

Philadelphia, PA

- Developing a hybrid sim-to-real learning framework for robotic manipulation of granular media that avoids the computational cost of high-fidelity particle simulators
- Built a Taichi-based Material Point Method simulator and graph-based state representation to generate training data; implemented a GNN surrogate that predicts surface-level material transitions from manipulator interactions
- Fine-tuning a residual model on real-world data from a Franka FR3 arm to close the sim-to-real gap

Apple: Apple Watch Product Design

Jun – Aug 2025

Product Design Intern

Cupertino, CA

- Designed and ran mechanical test protocols (Instron, Keyence) to evaluate durability of watch features
- Led design of a simplified drop-test model; redesigned internal flex PCBs for access without disassembly
- Released flex PCB and component designs to international vendors with EE and display teams
- Conducted competitor teardowns and analyzed product architecture

Projects

Quadcopter Trajectory Planning & Control / MEAM 6200 Advanced Robotics

Jan – May 2026

- Designed and tuned a PD position and attitude controller for the CrazyFlie quadcopter for stable 6-DOF flight
- Implemented A* and Dijkstra-based path planning to generate waypoint sequences through constrained environments (e.g., windows, slalom gates). Improved tracking smoothness and controller stability by pruning redundant nodes and fitting quintic polynomial splines to ensure C^2 continuity.
- Integrated planning and control pipelines in simulation and validated performance on physical hardware

Particle Filter SLAM on Real-World LiDAR Data / ESE 6500 Learning in Robotics

Jan – May 2026

- Built particle filter SLAM in Python using LiDAR and GPS/IMU data from the KITTI dataset: rigid-body motion model, log-odds occupancy mapping, LiDAR likelihood reweighting, and stratified resampling
- Validated on four urban, rural, and highway trajectories, comparing estimated paths against raw odometry

Block Pick-and-Place Competition / MEAM 5200 Intro to Robotics

Sep – Dec 2025

- Developed an algorithm to grasp blocks with AprilTags and stack them in a goal region using the Franka Emika Panda robotic arm, achieving 100% success rate in simulation and hardware.
- Utilized forward and inverse kinematics to determine the current and goal poses of the robot after implementing an existing perception function that returned the block position.
- Implemented denoising functions and sample-based motion planning algorithms (RRT, PRM) to avoid collisions; optimized algorithms to increase efficiency and speed.

ROBA Mechatronics Competition / MEAM 5100 Mechatronic Systems

Sep – Dec 2025

- Led mechanical design of a semi-autonomous robot using PID control, WiFi communication, and sensor fusion
- Designed robot sensor circuitry, soldered electronics and manufactured the robot body, including laser-cut acrylic chassis, 3D printed ballcaster and electronics casing.
- Wrote wall-following and coordinate-tracking routines in C/Arduino on ESP32-S2/C3 using Vive and time-of-flight sensing.